

Sec 4.1 – Using First and Second Derivatives

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

We have a second approach to determine local maxima and minima for a function, by considering the concavity of the function.

Theorem (The Second-Derivative Test for Local Maxima and Minima)

- If $f'(p) = 0$ and $f''(p) > 0$, then f has a local minimum at p .
- If $f'(p) = 0$ and $f''(p) < 0$, then f has a local maximum at p .
- If $f'(p) = 0$ and $f''(p) = 0$, then the test tells us nothing.

Use the second derivative test to determine whether the critical points of our earlier function $f(x) = x^5 - 10x^3 - 8$ are local maxima or minima. Recall that $f'(x) = 5x^4 - 30x^2$, $f''(x) = 20x^3 - 60x$ and the critical points of f are $x = 0, \pm\sqrt{6}$.

$$f(x) = x^5 - 10x^3 - 8$$

$$f'(x) = 5x^4 - 30x^2$$

$$f''(x) = 20x^3 - 60x$$

We had determined the critical points of f where $x = 0, \pm\sqrt{6}$.

- $f''(-\sqrt{6}) = 20(-\sqrt{6})^3 - 60(-\sqrt{6}) = -120\sqrt{6} + 60\sqrt{6} = -60\sqrt{6} < 0$, so this is a local maximum.
- $f''(0) = 0$, so we have no information (but we knew from before it was neither a maximum nor a minimum)
- $f''(\sqrt{6}) = 20(\sqrt{6})^3 - 60(\sqrt{6}) = 120\sqrt{6} - 60\sqrt{6} = 60\sqrt{6} > 0$, so this is a local minimum.

So if the function is concave up at a c.p., it is a local min, and if it is concave down at a c.p., it is a local max.

You should be able to use both the first and the second derivative tests.

An Inflection Point, i.p., is where the second derivative changes sign.

Definition

A point, p , at which the graph of a continuous function, f , changes concavity is called an **inflection point** of f .

Find the inflection points for our earlier function $f(x) = x^5 - 10x^3 - 8$. Recall that $f'(x) = 5x^4 - 30x^2$ and $f''(x) = 20x^3 - 60x$.

Solution:

$$f'(x) = 5x^4 - 30x^2$$

$$f''(x) = 20x^3 - 60x = 20x(x^2 - 3) = 20x(x - \sqrt{3})(x + \sqrt{3})$$

The zeros are $x = 0, \pm\sqrt{3}$. We need to determine if f changes concavity at each of these points.

- At $x = -\sqrt{3}$, $f''(x) < 0$ for $x < -\sqrt{3}$ and $f''(x) > 0$ for $-\sqrt{3} < x < 0$. Since concavity changes, this is an inflection point.
- At $x = 0$, $f''(x) > 0$ for $-\sqrt{3} < x < 0$ and $f''(x) < 0$ for $0 < x < \sqrt{3}$, so this is also an inflection point.
- At $x = \sqrt{3}$, $f''(x) < 0$ for $0 < x < \sqrt{3}$ and $f''(x) > 0$ for $x > \sqrt{3}$, so this is also an inflection point.

Partner up

Find the inflection points for $f(x) = x^3 - 9x^2 + 24x + 5$ and $g(x) = x^4$

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Solutions:

$$f'(x) = 3x^2 - 18x + 24 \quad f''(x) = 6x - 18$$

Then to find possible inflection points, we set $f''(x) = 6x - 18 = 0$, which implies that $x = 3$.

When $x < 3$, $f''(x) < 0$ and when $x > 3$, $f''(x) > 0$, so since concavity changes, this is an inflection point.

For the function $g(x) = x^4$,

$$f'(x) = 4x^3 \quad f''(x) = 12x^2$$

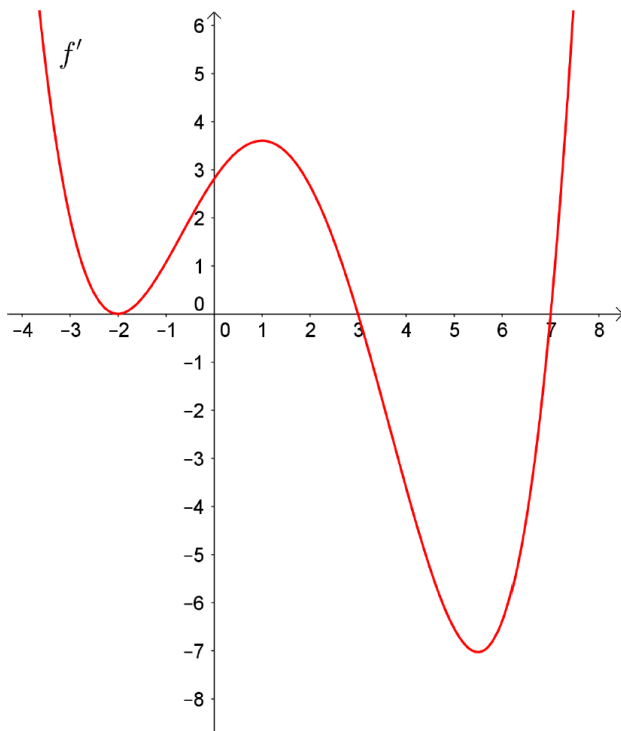
To find possible inflection points, we set $f''(x) = 12x^2 = 0$, which implies $x = 0$.

No matter what x is, $f''(x) = 12x^2$ is always non-negative (positive or zero) since we have an even power of x . This means that concavity does not change and this is not an inflection point.

Partner up

Below is the graph of the **derivative** f' .

Determine the critical points of f and whether each is a local maximum, local minimum, or neither.



Determine the inflection points of f .

Critical points are when $f'(x) = 0$. Looking at our graph, the critical points are at $x = -2, 3, 7$.

$x = -2$ is neither a local maximum or a local minimum - f' does not change sign at $x = -2$, but is positive on both sides, so this is just a level place of f between increasing sections.

$x = 3$ is a local maximum. Before $x = 3$, f is increasing since f' is positive and after $x = 3$, f is decreasing since f' is negative. Going up and then down means that the peak is at $x = 3$.

$x = 7$ is a local minimum since f' switches from negative to positive and hence f switches from decreasing to increasing, which makes $x = 7$ the lowest spot in its vicinity, a local maximum.

Inflection points occur when $f''(x) = 0$, and since f'' is the derivative of f' , this means that f' has a horizontal tangent. Therefore, the inflection points are at approximately $x = -2, 1, 5.5$, since f' changes from inc/dec to dec/inc at each.

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We can use all these ideas to sketch the graph of a function - even with graphing calculators, this can make sure we are paying attention to key points and not missing something off the screen.

Let $f(x) = e^{-x^2}$. Find all critical points, inflection points, and local extrema, and use this to sketch a graph of the function:

$$f'(x) = e^{-x^2}(-x^2)' = e^{-x^2}(-2x) = -2xe^{-x^2}$$

$$f''(x) = -2e^{-x^2} + 4x^2e^{-x^2} = -2e^{-x^2}(1 - 2x^2)$$

The critical points are when $f'(x) = 0$, which occurs when $x = 0$ (remember that the exponential function is *never* 0).

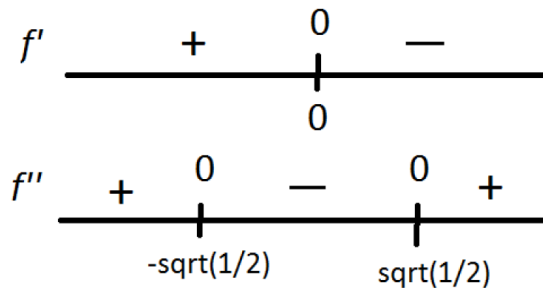
Using the second derivative test, $f''(0) = -2e^0(1 - 0) = -2 < 0$, so this is concave down and the critical value is a local maximum. (Alternately, you could see that $f'(x) > 0$ to the left of $x = 0$ and $f'(x) < 0$ to the right of $x = 0$.)

To find inflection points, we consider where $f''(x) = 0$. This means that $-2e^{-x^2} = 0$ or $1 - 2x^2 = 0$. The former cannot happen and the latter implies that $x = \pm\sqrt{1/2}$. We check these points:

- If we have an x -value less than $-\sqrt{1/2}$, say $x = -1$, we get $f''(-1) = -2e^{-1}(1 - 2) = 2e^{-1} > 0$.
- In $(-\sqrt{1/2}, \sqrt{1/2})$ we check $x = 0$: $f''(0) = -2e^0(1 - 0) = -2 < 0$.
- For an x -value greater than $\sqrt{1/2}$, we can choose $x = 1$ and $f''(1) = -2e^{-1}(1 - 2) = 2e^{-1} > 0$.

So we see that we change concavity at each of these points and $\pm\sqrt{1/2}$ are both inflection points.

We can record this information in this diagram:



Then we can work methodically to graph the function. It is frequently easier to work from left to right. For instance, when $x < -\sqrt{1/2}$, we know the graph is increasing and concave up and draw this portion.